CUSTOMER CHURN PREDICTION

Data Preprocessing, Feature Engineering,
and Classification Model Evaluation

Project type	Machine Learning / Classification
Dataset size	500 records and 9 columns
Target variable	Churn (binary classification)
Models evaluated	Logistic Regression and Random Forest

Prepared as a project report for academic review and submission.

1. Executive Summary

Customer churn prediction is a classification task that estimates whether a customer is likely to stop using a service. This project builds a reproducible preprocessing and modeling workflow using customer tenure, charges, contract type, payment method, paperless billing, and senior-citizen status. The workflow validates the source data, creates derived features, handles numerical and categorical variables, compares two classifiers, and evaluates performance with multiple metrics.

In the recorded evaluation, Random Forest achieved 99% accuracy and an F1 score of approximately 0.952, while Logistic Regression achieved 94% accuracy and an F1 score of approximately 0.786. These results come from one held-out split and should be treated as preliminary rather than proof of performance on future customers.

Project Objectives

- Inspect and validate the customer churn dataset.
- Apply suitable preprocessing to numerical, nominal, and ordered categorical features.
- Create additional features representing customer behavior.
- Detect potential outliers and scale numerical variables.
- Compare Logistic Regression and Random Forest using classification metrics.
- Document limitations and provide a repeatable workflow.

2. Dataset Description

The input file contains **500 rows** and **9 columns**. Validation recorded **0 missing cells** and **0 duplicate rows**. The target field is **Churn**, with 0 and 1 treated as the two classes in this project. The original data provider's documentation should be used to confirm the meaning of these labels.

Column	Role / interpretation
CustomerID	Identifier; excluded from model predictors
Tenure	Customer tenure, assumed measured in months
MonthlyCharges	Recurring monthly charge
TotalCharges	Accumulated charges
Contract	Contract category with an intended order
PaymentMethod	Nominal payment method
PaperlessBilling	Paperless billing status
SeniorCitizen	Customer indicator feature
Churn	Binary target label

Target distribution: Churn = 0: 447 records; Churn = 1: 53 records. The target is imbalanced, so accuracy alone is not sufficient for evaluating model usefulness.

3. Data Preprocessing

Data validation

Checked the expected schema, target labels, missingness, duplicate rows, and basic dataset characteristics.

Identifier exclusion

CustomerID is excluded because an identifier generally does not represent customer behavior and may encourage spurious patterns.

Missing-value handling

The source validation found no missing cells. Median imputation for numerical fields and most-frequent imputation for categorical fields are included for robustness.

Categorical encoding

One-hot encoding is used for nominal categories such as PaymentMethod and PaperlessBilling. Contract is ordinal-encoded as Month-to-month, One year, and Two year. Label encoding is demonstrated for target labels only.

Outlier analysis

IQR fences and absolute Z-scores above 3 are screening heuristics. The model pipeline clips numeric extremes using IQR bounds fitted on training data instead of automatically deleting records.

Feature scaling

Min-Max scaling is demonstrated separately. StandardScaler is used in the model pipeline. These methods are alternatives and are not applied sequentially.

Train/test split

An 80/20 stratified split with a fixed random seed preserves approximate class proportions. Learned preprocessing steps are fitted within the training pipeline to reduce leakage.

4. Feature Engineering

Nine derived features are defined to represent tenure, billing, contract commitment, and payment behavior. They are generated from predictor columns and do not use the target variable.

Feature	Definition / purpose
TenureYears	Tenure divided by 12, expressing tenure in years.
AvgHistoricalMonthlySpend	TotalCharges divided by Tenure; zero tenure is treated as missing.
MonthlyToTotalRatio	MonthlyCharges divided by (absolute TotalCharges + 1).
ContractCommitmentMonths	Maps Month-to-month, One year, and Two year to 1, 12, and 24 as a commitment proxy.
IsMonthToMonth	Indicator for month-to-month contracts.
HasElectronicCheck	Indicator for Electronic Check payment method.
PaperlessBillingFlag	Maps Yes/No paperless billing to 1/0.
TenureChargeInteraction	Product of Tenure and MonthlyCharges.
HighMonthlyChargesFlag	Indicator for monthly charges at or above the dataset median.

Feature-engineering caveat: HighMonthlyChargesFlag uses the full-dataset median in this teaching example, meaning the threshold is calculated before the train/test split. For a stricter production workflow, learn this threshold using training data only inside a custom transformer, or remove the feature. Other learned preprocessing steps are fitted through the training pipeline.

5. Modeling Approach

Two supervised classifiers are compared: Logistic Regression, an interpretable linear baseline, and Random Forest, an ensemble of decision trees capable of representing non-linear relationships and feature interactions. Both models are evaluated on the same held-out test set using accuracy, precision, recall, F1 score, and ROC-AUC. The notebook also produces confusion matrices and classification reports.

Recorded Model Results

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Random Forest	0.990	1.000	0.909	0.952	1.000
Logistic Regression	0.940	0.647	1.000	0.786	0.996

Interpretation: In this recorded split, Random Forest has the strongest F1 score and accuracy. Logistic Regression has higher recall, meaning it identified all positive churn cases in this test split, but its precision is lower, so more predicted churn cases were false positives. The preferred model depends on business costs, including the cost of missing a customer who will churn versus contacting a customer who will not churn.

6. Feature Analysis

The notebook includes exploratory feature ranking using numeric correlation with the target, mutual information, and Random Forest feature importance. These methods provide complementary views of possible predictive signal. Correlation primarily reflects linear association for numeric variables; mutual information can capture broader dependencies; tree-based importance reflects how the fitted forest uses features. Rankings are not causal explanations and can be unstable with small or imbalanced datasets.

7. Validation, Limitations, and Responsible Use

- The dataset contains 500 records and an imbalanced target, so a small number of test examples can materially change metrics.
- Reported scores come from one stratified train/test split. Use repeated cross-validation and a final untouched test set before relying on performance claims.
- Confirm definitions, units, and target-label meanings with the dataset source, particularly whether Tenure is measured in months.
- The HighMonthlyChargesFlag threshold is calculated before splitting in this educational version and should be moved into a training-fitted transformer for production.
- Model performance does not establish causation. Review feature effects and error types before using predictions to target retention offers.
- Do not expose real customer information in a public repository. Remove or anonymize identifiers and follow applicable privacy and data-handling rules.

8. Conclusion

This project demonstrates an end-to-end customer churn classification workflow, including dataset validation, feature engineering, categorical encoding, imputation, outlier handling, scaling, feature ranking, and model comparison. The recorded test results favor Random Forest on accuracy and F1 score for this particular split, while Logistic Regression shows higher recall. Recommended next steps are cross-validation, leakage-safe threshold engineering, error analysis, and validation on fresh data before deployment.

9. How to Reproduce the Results

- Install Python 3.10 or later.
- Install dependencies using: `pip install -r requirements.txt`.
- Keep `churn_data.csv` beside `churn_prediction_pipeline.ipynb`.
- Open the notebook in Jupyter and run all cells from top to bottom.

- Review generated metrics, plots, feature rankings, and validation assertions. Results may vary if the dataset or library versions change.

Project files: churn_prediction_pipeline.ipynb, churn_data.csv, preprocessing_report.md, feature_engineering_documentation.md, and requirements.txt.